

CodeNest

Home Interior Design, Furniture & Wall Painting Platform

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for **CodeNest**, a web-based platform dedicated to home interior design services, including interior design consultation, furniture selection and purchase, wall painting services, and related home decor offerings. This document is intended to serve as a shared reference between stakeholders, designers, and the development team throughout the design, development, testing, and deployment phases of the project.

1.2 Document Scope

CodeNest will allow homeowners and businesses to explore interior design ideas, browse furniture and wall-painting catalogs, request consultations with designers, book services, make payments online, and track the progress of their projects. The platform will also provide designers/vendors and administrators with dashboards to manage projects, orders, and content. This document covers the web application (desktop and mobile-responsive) and the associated backend services required to support it.

1.3 Intended Audience

- **Project Owner / Client** — to validate that requirements match business goals.
- **UI/UX Designers** — to guide wireframes and visual design decisions.
- **Developers and QA Engineers** — to understand what must be built and tested.
- **Project Managers** — to plan timelines, sprints, and resources.
- **Interior Designers / Vendors (end users)** — to understand platform capabilities.

1.4 Definitions, Acronyms and Abbreviations

Term	Definition
SRS	Software Requirements Specification
UI/UX	User Interface / User Experience
Admin	Administrator who manages the CodeNest platform
Vendor / Designer	A registered interior designer, painter, or furniture seller offering services on CodeNest
Client / Customer	An end user seeking interior design, furniture, or painting services
SKU	Stock Keeping Unit, a unique identifier for a furniture/product item
API	Application Programming Interface
RBAC	Role-Based Access Control

1.5 References

- IEEE Std 830-1998 — Recommended Practice for Software Requirements Specifications (used as a structural reference).
- CodeNest brand and business requirement discussions with stakeholders.

2. Overall Description

2.1 Product Perspective

CodeNest is a new, standalone web platform (with a mobile-responsive interface) that connects homeowners with interior designers, painters, and furniture vendors. It is not dependent on any legacy system but may integrate with third-party services such as payment gateways, cloud storage for images/3D models, email/SMS notification providers, and mapping services for on-site visits.

2.2 Product Features (Summary)

- Browsable portfolio and inspiration gallery of interior design, furniture, and wall painting work.
- Service catalog covering interior design consultation, furniture sales, and wall painting packages.
- Room visualizer / mood-board tool to preview colors, furniture layout, and themes.
- Online booking and consultation request system with scheduling.
- E-commerce style furniture catalog with cart, checkout, and order tracking.
- Secure online payments and invoicing.
- Designer/vendor dashboard for managing projects, quotes, and client communication.
- Admin panel for content, user, order, and vendor management.
- Customer reviews, ratings, and testimonials.
- Search and filter by room type, budget, style, color, and material.

2.3 User Classes and Characteristics

User Class	Description
Guest Visitor	Browses the website, views galleries and services without logging in.
Registered Customer	Creates an account, books consultations, purchases furniture, tracks orders and projects.
Designer / Vendor	Registered professional who lists services/products, manages bookings, and communicates with clients.
Administrator	Manages platform content, users, vendors, orders, and overall system configuration.

2.4 Operating Environment

- Client side: Modern web browsers (Chrome, Firefox, Safari, Edge) on desktop, tablet, and mobile devices.
- Server side: Cloud-hosted backend (e.g., Linux-based server environment) with a relational or NoSQL database.
- Responsive design supporting screen widths from mobile (360px) to large desktop displays.

2.5 Design and Implementation Constraints

- The system must be responsive and mobile-friendly, since a large share of users are expected to browse via mobile.
- Image and 3D content used in the visualizer must be optimized to keep page load times acceptable.

- All payment processing must comply with applicable data security standards (e.g., PCI-DSS via a certified payment gateway).
- The platform should be built using scalable, well-supported web technologies to allow future feature growth.

2.6 Assumptions and Dependencies

- Users have access to a stable internet connection and a modern web browser.
- Third-party services (payment gateway, SMS/email provider, maps) remain available and within their documented rate limits.
- Designers/vendors will provide accurate service descriptions, pricing, and availability.

3. System Features / Functional Requirements

Each subsection below describes a major functional area of CodeNest along with its key requirements, referenced using the format **FR-x.y** for traceability.

3.1 User Registration and Authentication

Allows visitors to create accounts and securely access personalized features.

- **FR-1.1:** The system shall allow users to register using email/phone and password, or via social login (Google/Facebook).
- **FR-1.2:** The system shall allow customers and designers/vendors to register under distinct account types.
- **FR-1.3:** The system shall support secure login, logout, and password reset via email/OTP verification.
- **FR-1.4:** The system shall allow users to update their profile information, including address and contact details.

3.2 Portfolio and Gallery Browsing

Showcases completed projects to inspire and inform prospective customers.

- **FR-2.1:** The system shall display a gallery of interior design, furniture, and wall painting projects with images.
- **FR-2.2:** The system shall allow filtering the gallery by room type (living room, bedroom, kitchen, etc.), style, and color theme.
- **FR-2.3:** The system shall allow users to view detailed project pages including description, designer, and materials used.

3.3 Service Catalog (Interior Design, Furniture, Wall Painting)

Presents the core service categories offered on CodeNest.

- **FR-3.1:** The system shall display service categories: Interior Design, Furniture, and Wall Painting, each with sub-services.
- **FR-3.2:** The system shall show package details, estimated pricing ranges, and service descriptions for each category.
- **FR-3.3:** The system shall allow customers to request a custom quote for a specific service.

3.4 Room / Design Visualizer

Helps customers preview design choices before committing.

- **FR-4.1:** The system shall provide a tool for users to preview wall paint colors on sample room images.
- **FR-4.2:** The system shall allow users to create simple mood boards by combining furniture, color, and decor items.
- **FR-4.3:** The system should allow saving and sharing of visualizer designs to a user's account (nice-to-have for future phases).

3.5 Booking and Consultation Requests

Enables customers to schedule designer visits or consultations.

- **FR-5.1:** The system shall allow customers to request a consultation by selecting a service, preferred date/time, and location.
- **FR-5.2:** The system shall notify the relevant designer/vendor of new booking requests.
- **FR-5.3:** The system shall allow designers to accept, reschedule, or decline booking requests.
- **FR-5.4:** The system shall allow customers to track the status of their booking (Pending, Confirmed, Completed, Cancelled).

3.6 Furniture and Product Catalog with Cart

Provides an e-commerce experience for furniture and decor items.

- **FR-6.1:** The system shall display furniture and decor products with images, price, dimensions, and material details.
- **FR-6.2:** The system shall allow users to add products to a cart, update quantities, and proceed to checkout.
- **FR-6.3:** The system shall allow users to view product availability (in stock / made to order).
- **FR-6.4:** The system shall recommend related or complementary products on product detail pages.

3.7 Order and Payment Management

Handles secure transactions and order lifecycle tracking.

- **FR-7.1:** The system shall support secure online payment via cards, UPI, net banking, and/or digital wallets.
- **FR-7.2:** The system shall generate an order confirmation and digital invoice after successful payment.
- **FR-7.3:** The system shall allow users to view order history and current order/delivery status.
- **FR-7.4:** The system shall support refund/cancellation workflows in accordance with the platform's policy.

3.8 Designer / Vendor Dashboard

Gives service providers tools to manage their business on the platform.

- **FR-8.1:** The system shall allow designers/vendors to create and manage their service and product listings.
- **FR-8.2:** The system shall allow designers/vendors to view and respond to booking requests and customer messages.
- **FR-8.3:** The system shall allow designers/vendors to update project status and upload progress photos.
- **FR-8.4:** The system shall provide designers/vendors a summary view of earnings and completed projects.

3.9 Admin Panel

Provides platform-wide oversight and management capabilities.

- **FR-9.1:** The system shall allow administrators to manage user, designer, and vendor accounts (approve, suspend, remove).
- **FR-9.2:** The system shall allow administrators to manage gallery content, service categories, and featured listings.

- **FR-9.3:** The system shall allow administrators to view platform-wide orders, bookings, and revenue reports.
- **FR-9.4:** The system shall allow administrators to moderate reviews and reported content.

3.10 Reviews, Ratings and Feedback

Builds trust through customer feedback on services and products.

- **FR-10.1:** The system shall allow customers to rate and review a designer/vendor or product after project/order completion.
- **FR-10.2:** The system shall display average ratings on designer/vendor profiles and product pages.
- **FR-10.3:** The system shall allow administrators to moderate or remove inappropriate reviews.

3.11 Notifications and Communication

Keeps users informed throughout their journey on the platform.

- **FR-11.1:** The system shall send email/SMS notifications for booking confirmations, order updates, and payment receipts.
- **FR-11.2:** The system shall provide an in-app messaging channel between customers and designers/vendors.
- **FR-11.3:** The system shall notify users of important status changes (e.g., booking accepted, order shipped).

3.12 Search and Filtering

Helps users quickly find relevant services, products, or inspiration.

- **FR-12.1:** The system shall provide a global search bar for services, products, and gallery items.
- **FR-12.2:** The system shall allow filtering by category, price range, room type, color, and material.
- **FR-12.3:** The system shall display relevant results ranked by relevance and, optionally, by rating or popularity.

4. External Interface Requirements

4.1 User Interfaces

- A responsive, visually appealing web interface reflecting a warm, modern home-decor aesthetic.
- Consistent navigation across Home, Gallery, Services (Interior Design / Furniture / Wall Painting), Shop, Bookings, and Account sections.
- Accessible design following common web accessibility practices (readable fonts, sufficient color contrast, alt text for images).

4.2 Hardware Interfaces

CodeNest does not require specialized hardware. It shall run on standard client devices (desktop, laptop, tablet, smartphone) with a supported web browser and internet connectivity.

4.3 Software Interfaces

- Payment Gateway API (for processing online payments securely).
- Email/SMS Service Provider API (for notifications and OTP verification).
- Cloud Storage Service (for storing gallery images, product photos, and visualizer assets).
- Maps/Location API (for scheduling on-site visits and validating service areas).

4.4 Communication Interfaces

The system shall communicate over standard HTTPS protocol to ensure encrypted data transmission between the client browser and backend services.

5. Non-Functional Requirements

5.1 Performance Requirements

- Standard web pages shall load within 3 seconds under normal network conditions.
- The visualizer tool shall render preview changes within 2 seconds of user interaction.
- The system shall support at least 1,000 concurrent users without significant performance degradation (initial target; scalable thereafter).

5.2 Security Requirements

- All sensitive data (passwords, payment details) shall be encrypted in transit (HTTPS/TLS) and at rest.
- The system shall implement Role-Based Access Control (RBAC) for Customer, Designer/Vendor, and Admin roles.
- The system shall never store raw card details; payments shall be processed via a PCI-DSS compliant gateway.
- The system shall log and monitor suspicious login attempts and support account lockout after repeated failures.

5.3 Usability Requirements

- The interface shall be intuitive enough for a first-time user to book a consultation within 3-4 steps.

- The platform shall be fully usable on mobile devices with touch-friendly navigation.
- Error messages shall be clear, specific, and guide users toward resolution.

5.4 Reliability and Availability

- The system shall target 99.5% uptime, excluding scheduled maintenance windows.
- The system shall include automated backups of critical data (orders, bookings, user accounts) on a daily basis.

5.5 Scalability and Maintainability

- The system architecture shall support horizontal scaling as user traffic grows.
- The codebase shall follow modular design principles to simplify future feature additions and maintenance.

5.6 Legal and Compliance Requirements

- The platform shall include a clear Privacy Policy and Terms of Service accessible to all users.
- The system shall comply with applicable data protection regulations relevant to the regions it operates in.
- Customer consent shall be obtained before sending marketing communications.

6. System Architecture Overview

CodeNest will follow a layered web application architecture consisting of a presentation layer (responsive web front-end), an application/business logic layer (API services handling bookings, orders, and user management), and a data layer (database for users, products, orders, and content). Third-party integrations (payment gateway, notifications, cloud storage, maps) will connect to the application layer via secure APIs.

Layer	Responsibility
Presentation Layer	Renders web pages, handles user interaction, responsive UI for all devices.
Application Layer	Business logic for bookings, orders, catalog, user roles, and notifications via REST/GraphQL APIs.
Data Layer	Stores users, products, orders, bookings, reviews, and gallery content.
Integration Layer	Connects to payment gateway, email/SMS provider, cloud storage, and maps services.

7. Data Requirements

Key data entities the system shall maintain include, but are not limited to:

Entity	Key Attributes
User	Name, email, phone, password (hashed), role, address
Designer/Vendor	Business name, portfolio, service areas, ratings, verification status
Service	Category (Interior Design/Furniture/Wall Painting), description, price range
Product	SKU, name, category, price, dimensions, material, stock status, images
Booking	Customer, service, designer, requested date/time, status
Order	Customer, items, total amount, payment status, delivery status
Review	Author, target (product/designer), rating, comment, date
Gallery Item	Title, category, images, associated designer, tags

8. Use Case Summary

Use Case	Primary Actor	Description
Register/Login	Customer, Designer	Create an account and securely access the platform.
Browse Gallery	Guest, Customer	View interior design, furniture, and painting project inspiration.
Request Consultation	Customer	Book a designer visit or consultation for a service.
Purchase Furniture	Customer	Browse catalog, add to cart, checkout, and pay.
Manage Listings	Designer/Vendor	Add/update service or product listings and pricing.
Manage Bookings	Designer/Vendor	Accept, reschedule, or decline consultation requests.
Leave Review	Customer	Rate and review a completed service or purchased product.
Manage Platform	Administrator	Oversee users, vendors, content, and reports across CodeNest.

9. Appendix

9.1 Future Enhancements (Out of Initial Scope)

- AI-based design recommendations tailored to uploaded room photos.
- Augmented Reality (AR) furniture placement preview.
- Loyalty/rewards program for repeat customers.
- Multi-language and multi-currency support for regional expansion.

9.2 Approval

This document requires review and sign-off from the project stakeholders before development begins. Any changes to scope after approval should be tracked via a formal change request process.